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From chaos to symbiosis: exploring adaptive co-evolution strategies for generative AI and research integrity systems

Wenqing Miao^{1†}, Huan Zang^{1†}, Qirui Liu^{2,3†}, Tianlei Zheng^{4,5}, Yan Zhou^{1,8}, Chunmei Liu¹, Na Yang⁵, Hengzhi Zhang¹, Yuwan Zhang¹, Ying Zhang¹, Shengli Li^{6,7*}, Shenyang Zhang^{1,8*} and Hao Zhang^{9,10*}

Abstract

Objective The information age has transformed technologies across disciplines. Generative artificial intelligence (GenAI), as an emerging technology, has integrated into scientific research. Recent studies identify GenAI-related scientific research integrity concerns. Using Complex Adaptive Systems (CAS) theory, this research examines risk factors and preventive measures for each agent within the scientific research integrity management system during GenAI adoption, providing new perspectives for integrity management.

Method This study applies CAS theory to analyze the scientific research integrity management system, identifying four core micro-level agents: researchers, research subjects, scientific research administrators, and academic publishing institutions. It examines macro-system complexity, agent adaptability, and the impact of agent interactions on the overall system. This framework enables analysis of GenAI's effects on the research integrity management system.

Results The scientific research integrity management system exhibits structural, hierarchical, and multidimensional complexities, with internal circulation of policy, funding, and information elements. In response to GenAI integration, four micro-level agents—researchers, research subjects, scientific research administrators, and academic publishing institutions—adapt their behaviors to systemic changes. Through these interactions, behavioral outcomes emerge at the macro level, driving evolution of the research integrity management system.

Conclusions Risks of scientific misconduct permeate the entire research process and require urgent governance. This study recommends that scientific research administrators promptly define applicable boundaries for GenAI in research to guide researchers. Concurrently, they should collaborate with relevant departments to establish

[†]Wenqing Miao, Huan Zang and Qirui Liu are co-first authors of this article.

*Correspondence:

Shengli Li
lishengli401@163.com
Shenyang Zhang
zhangshenyang0925@163.com
Hao Zhang
zhanghao@xzhmu.edu.cn

Full list of author information is available at the end of the article



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regulatory frameworks addressing potential GenAI-related misconduct. Academic publishing institutions must assume quality assurance responsibilities by strengthening review and disclosure protocols. Furthermore, research integrity considerations should be systematically integrated into GenAI's technological development and refinement.

Highlights

- Develops an analytical framework grounded in Complex Adaptive Systems (CAS) theory to map evolving interactions among researchers, research subjects, scientific research administrators, and academic publishing institutions within GenAI-integrated research ecosystems.
- Identifies self-reinforcing dynamics between GenAI adoption and integrity governance, wherein adaptive rule adjustments by agents reshape system-wide integrity thresholds.
- Proposes adaptive governance mechanisms that balance innovation safeguards with integrity guardrails, emphasizing context-sensitive policy calibration over universal solutions.

Keywords Generative artificial intelligence, Scientific research management, Research integrity

Background

In November 2022, OpenAI launched ChatGPT (Chat Generative Pre-trained Transformer), drawing widespread attention to generative artificial intelligence (GenAI) through its advanced natural language processing capabilities. On January 20, 2025, DeepSeek R1 was released with open-sourced model weights, becoming an accessible AI exemplar due to its low cost and open-source nature. These releases have catalyzed a global surge in GenAI adoption and rapid expansion of AI applications.

GenAI is a machine learning model based on neural networks with a pre-trained converter architecture. Trained on extensive datasets with numerous parameters, it offers high operational efficiency and interpretability [1]. GenAI's significant auxiliary capabilities attract widespread attention, including in scientific research. *Nature's* global AI survey reveals 31% of researchers use the technology, 17% daily, with 56% of postdoctoral researchers employing chatbots for coding tasks. The most common application (63%) is text optimization [2]. Obviously, GenAI can enhance scientific research efficiency but carries inherent risks. For example, technologies like ChatGPT and DeepSeek—used for article/image generation and research shortcuts—may fabricate references or provide false information [2–7]. Research integrity and authenticity are paramount. Any output failing strict scientific integrity standards should be excluded from the scientific body of knowledge [1]. Therefore, these practices likely raise scientific research integrity issues.

GenAI presents dual characteristics: high accessibility and utility for research, yet it exacerbates scientific integrity issues and challenges research integrity management. In China, GenAI remains at an early development stage, with limited academic research on its benefits and risks [8]. As a branch of scientific research management, research integrity management constitutes a complex system with multi-agent interactions. It exhibits complexity, interdisciplinarity, systematicity, and

comprehensiveness [9], requiring appropriate methodologies for in-depth analysis. Complex Adaptive Systems (CAS) theory applies to new challenges in social science and offers novel approaches to address complexity from technological advances and globalization [10]. Therefore, this study adopts CAS theory as its primary analytical framework.

CAS theory research focuses on micro-level agents and macro-level systems, emphasizing interactions among internal elements. This “bottom-up” approach frames the scientific research integrity management system as a macro-level complex system. We analyze micro-agents' responses to GenAI integration and identify four core agents—researchers, research subjects, scientific research administrators, and academic publishing institutions—as pivotal CAS units through the research lifecycle (knowledge production, governance, dissemination, application). Vulnerability transfer occurs among these agents under GenAI's impact. For example, researchers may abuse GenAI to generate text or falsify data (direct risk sources); falsified data may contaminate experimental datasets (indirect effects) [11]; traditional regulatory rules often fail to identify GenAI knowledge products (rule failure points); and resultant publications undermine peer review mechanisms (trust collapse points). In this process, the relationship between researchers and scientific research administrators forms a “rule-based game,” while the relationship between academic publishing institutions and research subjects constitutes a “quality verification network.” Through nonlinear interactions, these four types of entities promote a dynamic balance in the scientific research integrity system. This framework can be abstractly described as a complex adaptive system comprising multiple complex elements and adaptive agents.

In summary, this study focuses on critical research integrity issues in the context of GenAI impact. Using CAS theory as an analytical framework, it identifies risk points and proposes governance mechanisms for research integrity management.

Method

J. H. Holland proposed CAS theory in 1992 and published a seminal monograph on it in 1995. He articulated that a CAS comprises numerous adaptive agents, fundamentally characterized by macro-level system complexity and micro-level agent adaptability [12]. Holland conceptualizes agents as proactive elements, emphasizing that within a complex adaptive system, agents adhere to specific rules and dynamically adapt these rules based on accumulated experience. Consequently, he defines rules as strategic tools for describing agent behavior.

Specifically, CAS theory's complexity manifests in dual dimensions. Externally, it encompasses the dynamic architecture of the system's environment, where multiple factors (e.g., temporal and spatial dimensions) necessitate continuous adaptation to emerging challenges. Internally, it involves system complexity characterized by seven core elements: four properties – aggregation, nonlinearity, flow, and diversity; and three mechanisms – tags, internal models, and building blocks [13]. Concurrently, the self-adjustment and adaptation of internal agents foster novel interactive relationships and coupling phenomena, collectively defining the system's intrinsic complexity.

Adaptability refers to the adaptive feedback process whereby agents learn through interactions with other agents and their environment. These interactions generate novel rules while rendering some existing rules obsolete. Consequently, as agents modify their own behaviors, they propel the evolution of the entire complex adaptive system.

Building on this foundation, Holland proposed the ECHO model to simulate the adaptive feedback processes of agents within complex adaptive systems. Since then, CAS theory has gained multidisciplinary traction, with scholars progressively enriching its theoretical framework. Analyzing extensive CAS-related research, Lansing contends that this theory holds particular applicability in social science domains [14]. Therefore, this study adopts CAS theory, positioning the research integrity management system as a complex adaptive system. We propose a preliminary CAS-based model for this system (Fig. 1). The following section analyzes the theoretical model, examining how CAS theory aligns with research integrity management through “system complexity” and “agent adaptability” perspectives.

Results

System complexity analysis

The complexity of the external system for scientific research integrity management

As a branch of research management, scientific research integrity management operates in a complex social

environment influenced by policy, economic conditions, and geography. On one hand, innovative policies have significantly boosted researchers' scientific engagement while increasing managerial focus on research integrity. On the other, advances in generative AI like ChatGPT and DeepSeek are introducing new integrity challenges, prompting management institutions to update policy regulations and enhance personnel and technology. Meanwhile, policy and regulatory frameworks—including responsible entities, content, and constraints—vary across countries. Examples include Australia's Research Integrity Committee, the United States' Office of Research Integrity, Norway's National Committee for Research Ethics, and China's Ministry of Science and Technology. Furthermore, in free market economies, scientific research increasingly becomes commercialized, potentially undermining research integrity [15]. Koen Vermeir critiques the neoliberalization of scientific inquiry in his analysis, arguing that “the intrinsic uncertainty of the sciences is tamed and transformed into the predictability of useful results determined by a short-term vision” [15].

The complexity within the research management system

Regarding the internal entities of the research integrity management system, research integrity has increasingly gained policymakers' attention while simultaneously provoking intense debates within academia [16]. Some studies indicate that while abundant literature examines the impact of research integrity issues on scientific research, few empirical investigations focus on stakeholders beyond researchers—such as policymakers, funders, institutional leaders, editors, or publishers [17]. Research integrity should not be confined solely to the level of researchers' individual behavior, but should be considered within the broader scientific system [18]. The scientific research integrity management system inherently involves multiple agents, each governed by distinct behavioral rules and demand orientations. These agents actively adapt to systemic changes, which generate novel contradictions and challenges that subsequently impose new demands on the entire system. As exemplified by GenAI's impact, such technological shifts induce behavioral adaptations among agents within the system, ultimately transforming its overall structure.

Therefore, in research integrity governance, attributing integrity issues solely to researchers is inadequate. The analytical scope must expand to incorporate scientific research administrators' perspectives, encompassing multiple agents and implementing whole-process oversight to comprehensively assess integrity dynamics. This approach proves particularly crucial given GenAI's growing influence.

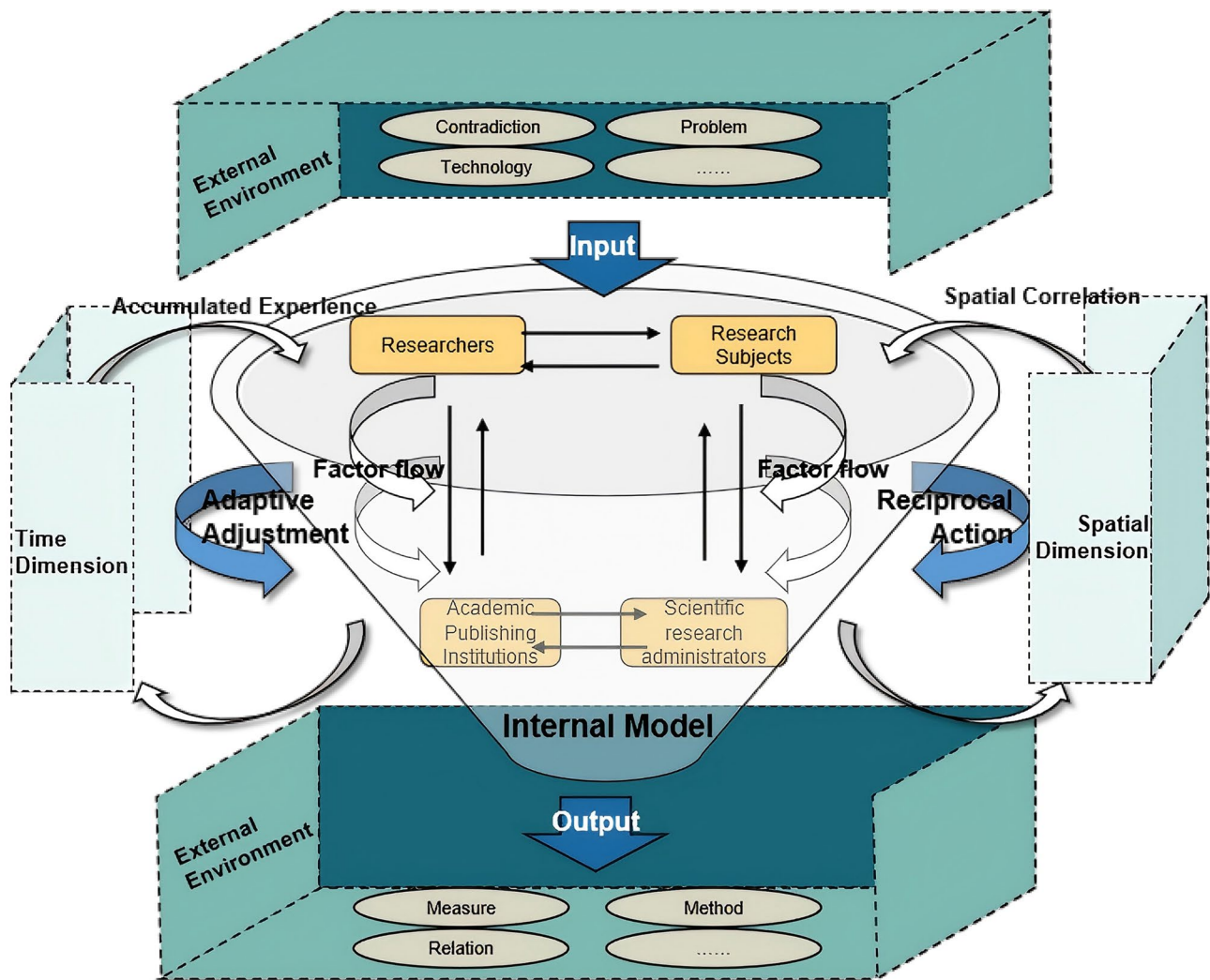


Fig. 1 Diagram of theoretical model of complex adaptive system. (Research integrity management system)

Analysis of agent adaptability

Within the research integrity management system's complex adaptive framework, stakeholders function as adaptive agents, while GenAI represents a novel element entering this ecosystem. Applying CAS theory enables deep analysis of how this influx affects both the macro-environment and micro-level agents, driving systemic evolution. Grounded in the "production-management-dissemination-application" research lifecycle framework and systematic literature analysis, this study identifies four core stakeholders: researchers (knowledge producers), research subjects (study targets), scientific research administrators (process regulators), and academic publishing institutions (dissemination facilitators).

Researchers: the "triggers" and "beneficiaries" of scientific research integrity risk

Researchers are simultaneously the focus of research integrity oversight, primary sources of integrity risk, and

core agents of scientific inquiry. Within their specialized fields, they independently conduct studies—designing experiments, generating data, and analyzing results to synthesize systematic findings for publication. Within the research integrity management system, researchers possess maximal autonomy and adaptability. As the core adaptive agents, they exhibit high sensitivity to environmental shifts and rapidly adjust behavioral strategies to maintain system alignment. When using GenAI, researchers—motivated by efficiency—may exploit the technology to seek research shortcuts for personal gain, thereby compromising integrity.

Research subjects: the stakeholders impacted by integrity breaches

Research subjects—whether human, animal, data, or events—are active participants in scientific inquiry. Under researcher observation or intervention, they contribute to data generation and serve as material carriers

that physically manifest the spirit of research integrity through standardized protocols. Take medical research as an example: the scientific research process is closely linked to relevant ethical and moral norms, and corresponding protections for research subjects' rights—such as informed consent and privacy—must be provided. In data analysis, a research subject can be a dataset. Researchers must ensure data authenticity and result reproducibility. However, data falsification or fabricated data will contaminate the dataset, causing scientific research integrity issues. The researcher's intervention may affect certain characteristics of the research subjects, and the results presented by the research subjects will in turn influence the researcher's decisions. Consequently, within the research integrity management system, the research subjects' behavioral pattern is essentially a response model that adapts to research needs under researcher guidance.

Scientific research administrators: the “controllers” of integrity risks

Scientific research administrators are primary operators of the research integrity management system, serving as observers, supervisors, and guides. Their work spans the entire research process and radiates across disciplines. Simultaneously, their performance critically enhances their institutions' research capacity. Within the system, administrators occupy core positions with network-style workflows extending to other agents. Vertically, they implement higher-level policies, communicate with researchers, and oversee research processes. Horizontally, they interface with other management systems—coordinating with finance personnel on funding allocation and legal staff on contract review. When systemic changes occur, administrators must keenly identify contradictions, respond rapidly, adjust behavioral patterns to establish new adaptive pathways, and continuously optimize management.

Academic publishing institutions: the “amplifiers” of integrity risks

Academic publishing institutions constitute the final stage from research output to public release, bearing responsibility for identifying and reviewing research results. If false or erroneous results enter academia through publishers, research integrity problems expand as citation chains spread exponentially—causing more researchers to form incorrect perceptions. Retraction exemplifies this issue: studies show 12.5% of retracted papers remain cited after 20 years [19]. Within the integrity management system, journals follow both unique publishing guidelines and unified academic norms, operating via established behavioral models. GenAI proliferation necessitates enhanced verification and detection

technology, exposing these models to increasingly active emerging challenges.

Complexity of the research integrity management system and adaptive nature of agents

Analysis of the above content confirms the research integrity management system's operation within a complex external environment, with inherently complex interactions among its internal agents. Horizontally, the system is shaped by temporally evolving factors; vertically, it interfaces with systems, regulations, institutions, and personnel. Spatially, distinct regional variations feature dynamic interflows of policies, funding, information, and personnel. Within this framework, each agent follows interest-oriented behavioral patterns. Consequently, the theoretical model in Fig. 1 is refined into Fig. 2 to reflect these systemic dynamics.

Discussions

Rule-based game: genai abuse vs. traditional scientific research governance

When GenAI integrates into scientific research, its influence pervades the entire research process. Initially, GenAI aids researchers in literature and data retrieval; however, studies indicate that such sourced materials lack reliability, and instances of GenAI fabricating literature have been identified [4]. Second, GenAI assists researchers in code writing and data analysis, undeniably enhancing research efficiency by automating auxiliary tasks. However, studies indicate critical concerns regarding its accuracy, authenticity, and ethical implications require assessment [3]. Third, GenAI's conversational interface enables autonomous article drafting per researcher specifications. Studies demonstrate that when provided specific requirements, GenAI generates complete research articles—including annotated sections like introduction, methods, results, and discussion—within approximately one hour. However, these outputs frequently contain substantive errors and inaccurate citations [20]. Additionally, scholars contend that GenAI-generated content derives from existing data and algorithms, thus cannot be recognized as the author's original work [6].

Fundamentally, GenAI integration transforms researchers' behavioral models. While enhancing research efficiency and quality, it risks enabling academic misconduct that cascades to other agents. Researchers gaining advantages through misconduct diminish peers' innovative output and motivation, increase publishers' challenges in identifying original content, and impose additional burdens on scientific research administrators.

Consequently, traditional research governance rules prove inadequate in countering emerging research integrity challenges. On one hand, traditional detection methods—including plagiarism checks and statistical

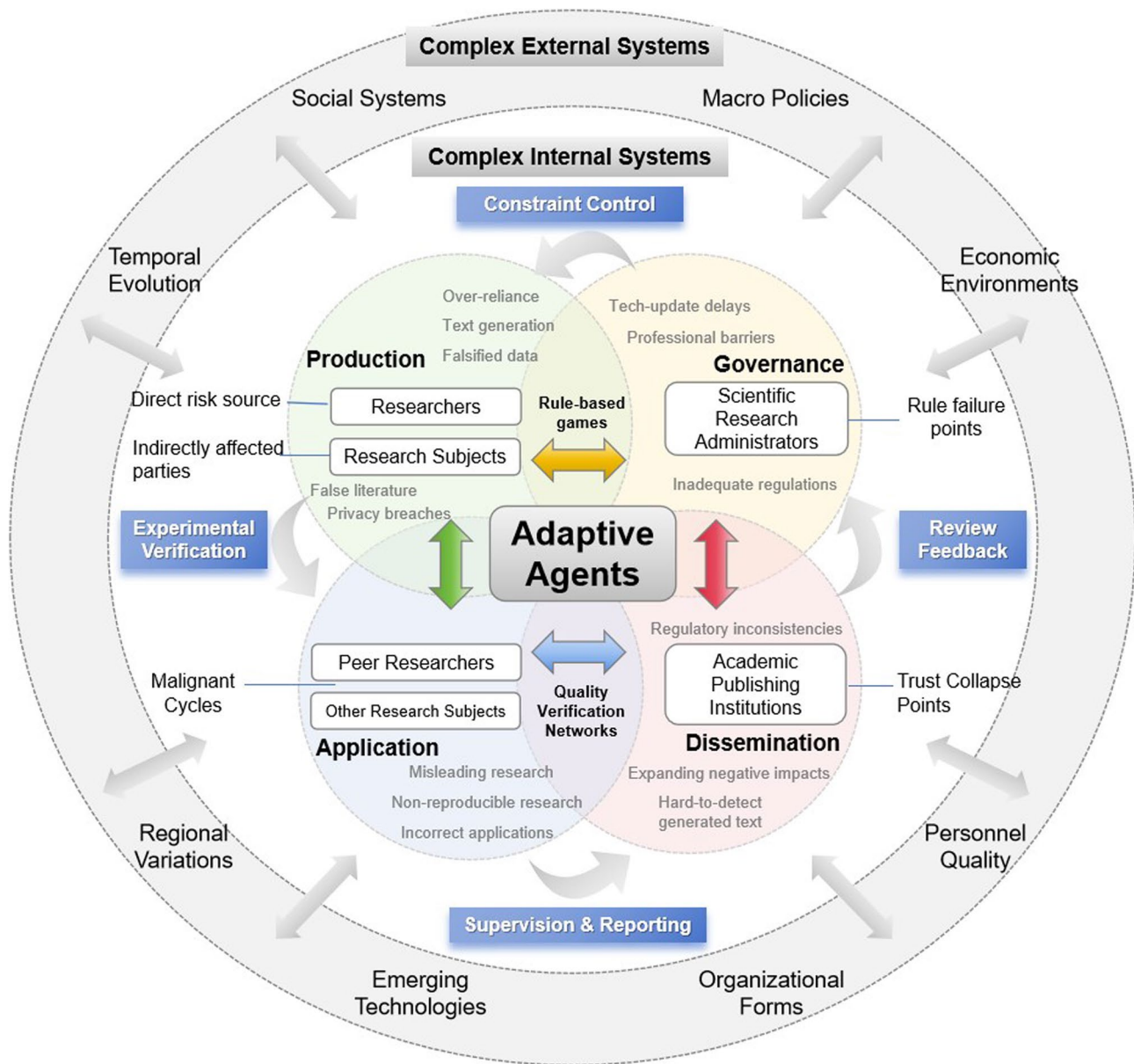


Fig. 2 Complex Adaptive Systems Framework for Research Integrity Management

analysis—fail to accurately identify GenAI fraud patterns, particularly hybrid outputs combining authentic local data with AI-generated conclusions [21]. On the other hand, GenAI involvement has caused blurring of the responsibility chain in scientific research. When academic misconduct accusations occur, it is difficult to accurately determine whether the fault lies with the researcher or the GenAI. Therefore, scientific research administrators' management rules require timely updates, such as: releasing GenAI usage guidelines, requiring authors to disclose its use, and prohibiting direct use of AI-generated references [22]. Furthermore, detection technologies require timely updates. For example, programmers developing large language models can

embed specific identifiers into algorithms, enabling easier recognition of AI-generated content. However, studies indicate that advancing detection technologies may inadvertently refine AI writing techniques, casting uncertainty on the development of accurate and reliable detection tools [23]. Therefore, this study states that a hybrid decision-making mechanism is needed – for example, adding human supervision weights during AI detection – to prevent algorithms from overcontrolling the academic ecosystem.

Supported by advanced AI detection technology and strict management rules, systems can form self-adaptive networks through nonlinear interactions among agents, reducing the incidence of research integrity issues and

maintaining the dynamic balance of the research integrity management system.

Synergistic improvement of researchers' submission strategies and publishing institutions' review rules

In the scientific research ecosystem, the relationship between academic publishing institutions and research subjects is mediated by researchers. Through nonlinear interaction and restrictive relationships, they jointly establish a quality verification network. This network ensures both the authenticity and reproducibility of research results and, via technological upgrades and institutional regulation design by scientific research administrators, forms closed-loop supervision of scientific research integrity.

On one hand, amid widespread GenAI adoption, researchers may falsify experimental data using this technology, generate research outputs, and submit them to academic publishing institutions. This severely compromises the authenticity and reproducibility of research subjects. If undetected by publications, false research spreads exponentially through citation chains. Therefore, as the ultimate gatekeepers of academic quality, academic publishing institutions must promptly enhance their original rules and guidelines, enforcing rigorous quality verification through updated submission rules, review processes, and technical tools [24]. Notably, several leading international journals have issued guidelines on GenAI use. For example, *Science* has shifted its policy: initially prohibiting ChatGPT-generated papers, disallowing AI-created images in submissions, and banning ChatGPT authorship, it now permits authors to use this technology for “the writing or presentation of the manuscript” under full disclosure [25]; *Nature* adopts a more conservative approach, permitting specification of ChatGPT's contributions in designated sections but explicitly barring its authorship [26]; similarly, *Elsevier* and *JAMA Network* have implemented comparable provisions [27].

On the other hand, approximately 47% of research integrity issues originate in the peer review process, primarily due to false peer reviews. Consequently, GenAI involvement in academic evaluation warrants attention [19]. Using GenAI in peer review can save time and improve efficiency. However, this approach may cause inconsistent review quality. Studies testing GenAI's content detection capabilities reveal it fails to grasp textual logical structure, continuity, and completeness, detecting only typographical and grammatical errors [28]. When GenAI serves as a peer reviewer, it may overlook profound logical flaws in human-authored manuscripts. Conversely, as an author, GenAI-generated papers challenge human reviewers' detection capabilities. Current detection tools demonstrate limited accuracy and reliability [29]. As stated in *Science*, a recent study tested

experts' ability to distinguish ChatGPT-generated abstracts from original research content, finding they identified only 63% of AI-generated abstracts [25]. Overreliance on GenAI in peer review directly compromises review quality, triggers academic confidence crises, and undermines research integrity systems. *Science* explicitly prohibits AI use in peer review due to manuscript confidentiality requirements [30]. Reviewers and authors must disclose their technology use and are responsible for data security, confidentiality, and report accuracy, logic, and originality [31].

Therefore, amid increasing GenAI adoption, academic publishing institutions must implement new adaptations: extending beyond author and manuscript oversight to regulate technology usage by editors and reviewers. These institutions should establish clear standards and enhance management protocols—including GenAI pre-screening and rigorous human expert verification—to develop efficient human-AI collaborative review networks. This approach reduces research integrity issues while ensuring research authenticity and reliability, effectively regulating researchers' behavioral patterns. Such measures safeguard research subjects' interests and enable collaboration with scientific research administrators to stabilize the integrity management system.

Suggestions for adjustment of scientific research administrators

GenAI, as a new technology, flows into the complex scientific research integrity management system, intensifying its complexity and enhancing the availability of system environmental information. Simultaneously, GenAI disrupts agent behavioral models and triggers updates to their adaptive paths. To maintain systemic stability and balance, appropriate regulatory constraints must be established. Based on this analysis, this study proposes the following suggestions for optimizing GenAI applications from a research integrity management perspective.

First, scientific research administrators must strengthen guidance for researchers on proper GenAI application. They should publicize GenAI's advantages and limitations through diverse media channels, enabling researchers to understand its drawbacks—particularly inaccurate data collection and retrieval—and avoid overreliance. Additionally, administrators could establish a research integrity platform using information technology, featuring a dedicated GenAI section to periodically expose global integrity violations.

Second, scientific research administrators must clarify boundaries for GenAI use in research and its relationship with research integrity issues. This process comprises four phases:

Phase 1: Issue Identification.

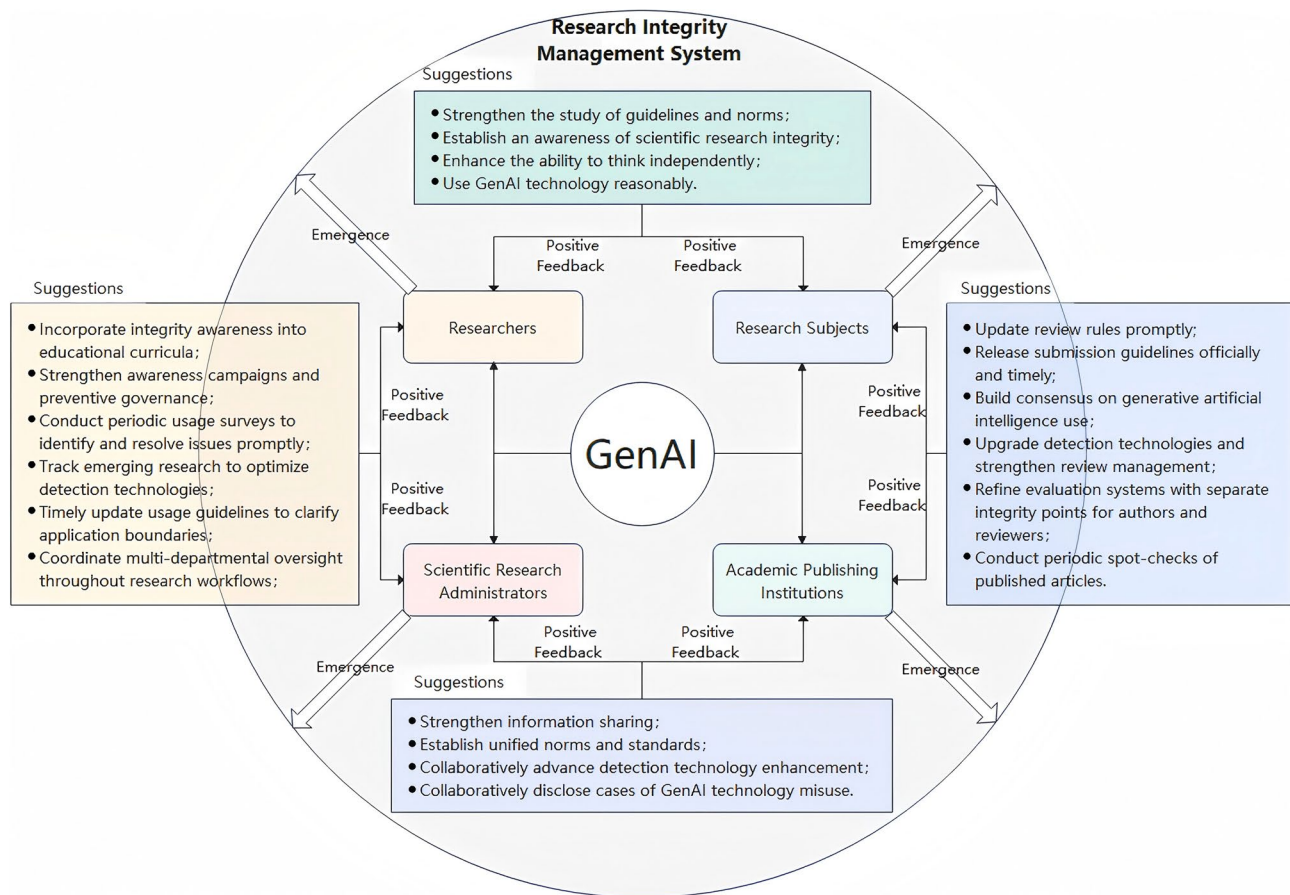


Fig. 3 Summary of Recommendations for Key Agents

After GenAI's public release, summarize and analyze research integrity problems. Categorize issues by nature, relevant stages, responsible agents, and subject types to identify the most frequent conflicts between GenAI and research integrity.

Phase 2: Targeted Analysis.

Extract high-incidence problems from Phase 1. Solicit expert opinions across disciplines through questionnaires, qualitative interviews, or the Delphi method to develop solutions.

Phase 3: Guideline Development.

Synthesize Phase 2 data to establish a GenAI-research integrity reference database. Analyze the technology's application boundaries in research processes and draft usage guidelines.

Phase 4: Pilot Implementation.

Test guidelines in limited scope. Gather feedback from researchers, research subjects, and scientific research administrators via periodic surveys to refine guidelines iteratively.

Third, scientific research administrators must collaborate with relevant agencies to establish timely regulatory frameworks [32]. For example, hospital research administrators could partner with health commissions, education

departments, and intellectual property offices to develop GenAI research guidelines, leveraging institutional authority to ensure norm efficacy and enable independent researcher compliance. Additionally, horizontally, hospital research administrators should coordinate with legal affairs, ethics offices, information departments, equipment divisions, and ethics committees to formulate institutional codes of conduct and implement multi-link reviews, thereby restricting researchers' GenAI use.

Based on the above analysis, this study proposes agent-specific recommendations for different agents (Fig. 3). Through positive feedback effects on each agent, their behavioral models are regulated, ultimately promoting macro-level stability in the evolution of the research integrity management system.

Conclusions

GenAI holds significant future potential in scientific research. Currently, it plays critical roles across research stages: literature review in early phases, experimental design and problem-solving in mid-phases, and manuscript writing/editing in late phases. While GenAI brings substantial convenience and possibilities, it also poses challenges including data privacy, ethical concerns,

content accuracy, and model interpretability. Therefore, GenAI should serve a supportive role in research processes. Researchers must avoid over-reliance and approach GenAI development with open yet cautious attitudes. Additionally, scientific research administrators and policymakers must clarify emerging technologies' impacts and depths on research, refining regulatory systems for timely constraints.

Limitations

This study focuses on qualitative analysis of GenAI's impact on scientific research integrity, constructing a theoretical framework for the research integrity management complex adaptive system. However, it lacks quantitative analysis support. Future research will survey researchers' understanding and usage of this technology or employ computer simulation to model potential research integrity trends under unregulated conditions.

Abbreviations

CAS Complex Adaptive Systems
GenAI Generative Artificial Intelligence

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Authors' contributions

WQM and HZ conducted literature search, planned the study and interpretation and drafted the manuscript. QRL, TLZ and SYZ, conducted frame design, feasibility analysis and helped conceptualize the project. YZ, CML, and NY contributed to the literature search and analysis. HZZ, YWZ and YZ conducted the format review. SYZ and SLL revised the manuscript and gave critical feedback. HZ conducted quality control and review of manuscripts. All authors reviewed the manuscript.

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Data availability

We do not analyse or generate any datasets, because our work proceeds within a theoretical approach.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Author details

¹Department of Science and Technology, The Affiliated Hospital of Xuzhou Medical University, Xuzhou, Jiangsu 221006, People's Republic of China

²Propaganda Department, The Affiliated Hospital of Xuzhou Medical University, Xuzhou, Jiangsu 221006, People's Republic of China

³School of Public Administration and Society, Jiangsu Normal University, Xuzhou, Jiangsu 221116, People's Republic of China

⁴School of Information and Control Engineering, China University of Mining and Technology, Xuzhou, Jiangsu 221116, People's Republic of China

⁵Artificial Intelligence Research Group, Medical Equipment Management Department, The Affiliated Hospital of Xuzhou Medical University, Xuzhou, Jiangsu 221006, People's Republic of China

⁶Clinical Research Institute, The Affiliated Hospital of Xuzhou Medical University, Xuzhou, Jiangsu 221006, People's Republic of China

⁷School of Computer Science and Technology, China University of Mining and Technology, Xuzhou, Jiangsu 221116, People's Republic of China

⁸Department of Neurology, the Affiliated Hospital of Xuzhou Medical University, Xuzhou, Jiangsu 221006, People's Republic of China

⁹President's Office, the Affiliated Hospital of Xuzhou Medical University, Xuzhou, Jiangsu 221006, People's Republic of China

¹⁰Department of Thoracic Surgery, The Affiliated Hospital of Xuzhou Medical University, Xuzhou, Jiangsu 221006, People's Republic of China

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